

CPE 381 CLASSWORK 08: CONTINUOUS-TIME SIGNALS, LINEARITY & SUPERPOSITION

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 16, 2026 (In-Class)

1. Check if the signal $x(t) = e^{-at}u(t)$, $a > 0$ is an energy signal. Note: a signal is energy signal if its energy is finite. **(10 Points)**

Answer

$$\begin{aligned}
 E &= \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |e^{-at}u(t)|^2 dt = \int_0^{\infty} e^{-2at} dt \\
 &= -\frac{1}{2a} \left[e^{-2at} \right]_0^{\infty} = -\frac{1}{2a} [0 - 1] = \frac{1}{2a} < \infty
 \end{aligned}$$

Clearly, the signal has finite energy. Hence, the signal is an energy signal.

2. For a system defined by $y(t) = T[x(t)]$, determine if it is additive where the system T is specified by $y(t) = x(t) + 3$. **(5 Points)**

Answer**Not additive.**

$$\text{Check: } T[x_1(t) + x_2(t)] = (x_1(t) + x_2(t)) + 3$$

$$T[x_1(t)] + T[x_2(t)] = (x_1(t) + 3) + (x_2(t) + 3) = x_1(t) + x_2(t) + 6$$

Since $3 \neq 6$, the property fails.

3. Is $y(t) = x(t) + 3$ homogenous? **(5 Points)**

Answer**Not homogeneous.**

$$\text{Check: } T[\alpha x(t)] = \alpha x(t) + 3$$

$$\alpha T[x(t)] = \alpha(x(t) + 3) = \alpha x(t) + 3\alpha$$

Since $3 \neq 3\alpha$ for $\alpha \neq 1$, the property fails.